

Thermal Chamber Validation & Telemetry Log

Fast-Charging Uniformity, Climatic Range, and Thermal Runaway Alert Verification

Facility Item	Specification Parameter	Facility Item	Specification Parameter
Test Laboratory:	Weiss Environmental Testing Center (Stuttgart)	Test Chamber:	Weiss ClimeEvent C/1000/70/3
Coolant Fluid:	Ethylene Glycol / Water 50:50 Premix	Flow Rate:	12.5 Liters/min @ 3.0 bar pressure
Battery Pack:	BMS-X800 Pack SN: 800-042 (800V Architecture)	Test Protocol:	ECE-R100 / ISO 26262 Thermal Safety

Executive Test Summary & Scope:

This engineering test report documents verification of the thermal management subsystem for the BMS-X800. Testing includes: (1) 3C DC fast-charge cell temperature gradient monitoring, (2) Extreme ambient climatic temperature qualification (-40°C to +85°C), and (3) Gas venting / thermal runaway CAN-FD broadcast latency. Detailed sensor telemetry plots and non-conformance reports are appended below.

1.0 Fast-Charge Cell-to-Cell Temperature Uniformity (REQ-BMS-004)

Test Protocol: 3C DC Fast Charging (120 kW constant current phase from 10% to 80% SoC).

Requirement REQ-BMS-004 Criterion: Maximum cell-to-cell temperature differential $\Delta T \leq 5.0$ deg C.

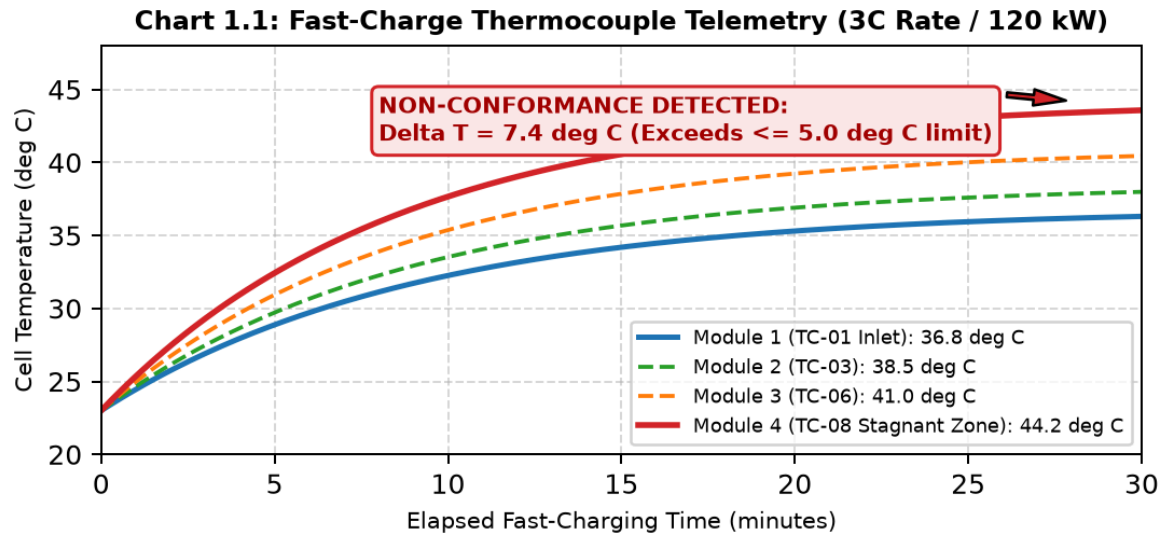


Table 1.2: End-of-Charge Thermocouple Peak Temperature Matrix (80% SoC)

Battery Module	Sensor ID	Peak Temp	Rise from 23C	Compliance Evaluation
Module 1 (Inlet)	TC-01	36.8 deg C	+13.8 deg C	Baseline reference (coolest module)
Module 2	TC-03	38.5 deg C	+15.5 deg C	Nominal coolant heat uptake
Module 3	TC-06	41.0 deg C	+18.0 deg C	Acceptable thermal spread
Module 4 (Hotspot)	TC-08	44.2 deg C	+21.2 deg C	NON-CONFORMING: Delta T = 7.4 deg C (Threshold <= 5.0 deg C)

2.0 Extreme Climatic Temperature Range Validation (REQ-BMS-005)

Test Protocol: 48-hour climatic profile soak and functional checkout.
Requirement REQ-BMS-005 Criterion: Full functionality across -40.0 deg C to +85.0 deg C range.

Environmental Condition	Chamber Temp	Dwell Time	Observed Functional State	Evaluation
High-Temperature Soak	+85.0 deg C	12 hours	Normal CAN-FD telemetry	PASS
Upper Operating Plateau	+70.0 deg C	12 hours	Pre-charge functional	PASS
Room Ambient Baseline	+23.0 deg C	8 hours	All cell channels calibrated	PASS
Intermediate Cold Dwell	-20.0 deg C	12 hours	Boot successful, monitoring active	PASS
Extreme Sub-Zero Soak	-40.0 deg C	0 hours (NOT RUN)	NOT EVALUATED (Chamber limit)	TEST GAP / PARTIAL

Section 2.3 Environmental Test Deviation & Traceability Note:

DEVIATION REPORT DEV-ENV-2026-088:
During the environmental test execution, the secondary cryogenic cascade compressor experienced low pressure, limiting minimum chamber air temperature to -20.0 deg C. While continuous operation was successfully demonstrated from -20.0 deg C up to +85.0 deg C, the mandatory extreme cold qualification at -40.0 deg C required by REQ-BMS-005 was NOT completed. Evidence for REQ-BMS-005 is therefore PARTIAL and requires additional validation.

3.0 Thermal Runaway Warning Broadcast Latency (REQ-BMS-006)

Test Protocol: Gas sensor injection trigger (simulated cell vent gas at Module 6).
Requirement REQ-BMS-006 Criterion: CAN-FD warning broadcast latency must be strictly < 100 ms.

Test Trial	Trigger Mechanism	Transmitted Frame	Measured Latency	Evaluation
Event Trial 1 (Module 6 Vent)	Gas Sensor Trip	CAN-FD ID 0x018	48 milliseconds	PASS (< 100 ms)
Event Trial 2 (Module 2 Vent)	Gas Sensor Trip	CAN-FD ID 0x018	46 milliseconds	PASS (< 100 ms)
Event Trial 3 (Thermal Spike >1C/s) dT/dt Threshold		CAN-FD ID 0x018	52 milliseconds	PASS (< 100 ms)
Event Trial 4 (Thermal Spike >1C/s) dT/dt Threshold		CAN-FD ID 0x018	49 milliseconds	PASS (< 100 ms)

Section 3.1 Runaway Latency Conclusion:
Across all four simulated thermal runaway venting and rapid temperature spike trials, the maximum recorded latency from sensor trigger to vehicle CAN-FD bus frame broadcast was 52 ms, with an average of 48.7 ms. All values satisfy the REQ-BMS-006 maximum latency limit of 100 ms, confirming compliance.